

Automatic Semantic Segmentation of the Lumbar Spine: Clinical Applicability in a Multi-Parametric and Multi-Center Study on Magnetic Resonance Images

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Inventory Category:	Segmentation / localization
Comparability / Priority:	Supporting methodology Priority: Critical

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Multicenter segmentation Multi-parametric, multi-center lumbar MRI
Dataset & Cohort:	None (Sample: None)
Disease / Target:	Lumbar anatomy
Model / AI Method:	CNN semantic segmentation
Evaluation Metrics:	None
Key Finding / Relevance:	None
Thesis Context (Ch 2):	Direct evidence that segmentation generalization across centers/protocols is a research problem.

Abstract / Summary

One of the major difficulties in medical image segmentation is the high variability of these images, which is caused by their origin (multi-centre), the acquisition protocols (multi-parametric), as well as the variability of human anatomy, the severity of the illness, the effect of age and gender, among others. The problem addressed in this work is the automatic semantic segmentation of lumbar spine Magnetic Resonance images using convolutional neural networks. The purpose is to assign a class label to each pixel of an image. Classes were defined by radiologists and correspond to different structural elements like vertebrae, intervertebral discs, nerves, blood vessels, and other tissues. The proposed network topologies are variants of the U-Net architecture. Several complementary blocks were used to define the variants: Three types of convolutional blocks, spatial attention models, deep supervision and multilevel feature extractor. This document describes the topologies and analyses the results of the neural network designs that obtained the most accurate segmentations. Several of the proposed designs outperform the standard U-Net used as baseline, especially when used in ensembles where the output of multiple neural networks is combined according to different strategies.